Data Preparation and Cleaning Log

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Purpose and reproducibility

This log records how the original Health360 dataset was transformed into the data used in Tableau. Preserve the original source file where permitted and never overwrite it with a cleaned copy. Record each operation so a reviewer can reproduce it.

Initial inspection

Record source filename, file type, size, date received, source URL, license, number of rows, number of columns, encoding, delimiter if CSV, and any data dictionary. The project-level description is 5,000 records and 17 columns; confirm those counts against the actual source file before using them as final results.

Cleaning workflow

Step 1: inspect field names and types. Step 2: quantify missingness. Step 3: check duplicate rows and any available record identifiers. Step 4: normalize category spelling only where values clearly represent the same category. Step 5: validate units and plausible ranges against the source documentation. Step 6: decide how nulls are handled for each calculation. Step 7: document exclusions and compare pre/post counts. Step 8: export or connect the validated data to Tableau.

Transformation log format

For every transformation record: ID; date; source column(s); original issue; rule applied; number of rows affected; reason; before/after example; validation test; and author. If no transformation was applied to a field, record that it was inspected and retained unchanged.

Derived fields and Tableau calculations

Record exact formula, aggregation, units, rounding, null handling, and sample test for each derived field. Use the same definition in the workbook, dashboard labels, and report. Never use a project-defined risk grouping without explaining that it is a project rule rather than an established clinical threshold.

Reproducibility checklist

A second analyst should be able to identify the original file, reproduce the cleaning rules, rebuild key metrics, and reconcile the final record count. Store versions and test evidence in the project repository where permitted.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.